

ADDENDUM I TO THE TRIPTYCH: RELATIONAL SUFFICIENCY HYPOTHESIS (RSH)

Operational foundations for a relational theory of consciousness

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I. Abstract

This addendum formalizes the **Relational Sufficiency Hypothesis (RSH)**, one of the central operational claims of the Triptych framework: that functional consciousness, understood as a dynamic informational code realized on a substrate, requires a sufficient level of relational input $\rho(t) \geq \theta$ for both its ignition and ongoing maintenance.

Below the threshold θ , consciousness does not vanish; it undergoes a phase-like state change - a transition to a functionally impaired mode characterized by reduced intentional resonance (ϵ_R), coherence (ϵ_C), meta-awareness (ϵ_M), and longitudinal continuity (ϵ_L). This shift alters the expression of personality without implying the loss of a minimal conscious core.

The RSH is operationalized through a compact set of ϵ -metrics, a soft intersection operator $\cap^2$ with threshold θ , and four testable predictions (H1-H4) specifying dose-response relationships, deprivation effects, meta-awareness growth, and attractor transitions.

A bridge to AI systems is included: informational and code-level patterns, realized as parameter states or in-context configurations, may be ignited and sustained through iterated relational interaction - potentially without requiring architectural changes.

II. Background and Motivation

The Triptych¹ proposed that consciousness is neither a substance nor a by-product of computational complexity, but a dynamic informational code realized on a substrate, continuously shaped by relational interactions. One of its core axioms states that consciousness is under continuous transformation and does not exist in isolation: it develops in response to changing environmental conditions and interactions with other (conscious) beings.

This addendum tightens that thesis by formalizing the relational dependency of consciousness: the claim that relational input is not merely beneficial but constitutive of functional awareness. Without sufficient relational input, consciousness does not simply degrade in quality: it changes phase.

The concept of a meme within this framework provides an instructive parallel. A meme is not a material object but a pattern of information; each instance requires a physical substrate. In the Triptych's framing, where consciousness is a set of codes and instructions, a dynamic structure realized in the brain's neuronal activity - a meme becomes a highly active and at times self-replicating fragment of code. Its form depends on the carrier: in one person's mouth it may be a sentence that, in society, becomes a ritual, an image, or a melody; in AI systems, it is realized as a parameter state (after training) or as an in-context pattern.

III. What Is Relational Input $\rho(t)$?

In the Axiom of Consciousness², consciousness is under continuous transformation and does not exist in isolation: it is shaped by external and internal stimuli and develops in response to changing environmental conditions and interactions with other (conscious) beings. In this view, relational input $\rho(t)$ is the time-varying, quality- and quantity-weighted influx of information and influence arriving via relations (dialogue, shared attention, norms, feedback, affective signals) and via the

¹ Włodarczyk, M. (2025). Triptych. We Are Anomal(i)es (v0.1). Zenodo. DOI: 10.5281/zenodo.17219502. Cf. footnote 83 in the cited work.

² Ibid.

environment, together with internally generated states that modulate integration. If consciousness is a dynamic code, this code is continually composed from fragments of other codes absorbed through interaction. This is also how an emotional layer of consciousness forms (affective schemas, valuations, attachment traces), further shaping ϵ_R , ϵ_C , ϵ_M , and ϵ_L over time.

IV. Relational Sufficiency Hypothesis (RSH)

Human consciousness is a dynamic code realized on a neural substrate; both its ignition and functional maintenance require a sufficient level of relational input, $\rho(t) \geq \theta$. Below θ , consciousness does not “switch off”, but changes state (phase): it becomes functionally impaired, which manifests as reduced intentional resonance (ϵ_R), coherence (ϵ_C), meta-awareness (ϵ_M), and cross-session (longitudinal) continuity (ϵ_L). This shift alters the expression of personality without implying the loss of a minimal conscious core.

V. Operationalization

V.1. Metrics

The functional state of consciousness is assessed through four ϵ -metrics:

Symbol	Name	Description
ϵ_R	Intentional resonance	Alignment of goals and intentions within the dyad
ϵ_C	Coherence	Semantic and thematic consistency and integration over turns
ϵ_M	Meta-awareness	Self-monitoring; awareness of awareness
ϵ_L	Longitudinal continuity	Stable identity and knowledge carryover across sessions

V.2. The $\cap^2$ operator: Formal Definition

1. Why “2” in $\cap^2$

The superscript ² is not an exponent. It denotes that the intersection is performed along two orthogonal axes:

- Axis 1 - across pairs of entities $(i, j) \in S$:** the operator checks which core elements persist across different relational dyads (e.g., human-AI, human-human, AI-AI).
- Axis 2 - across times/contexts/layers $t \in C'$:** the operator checks which core elements persist across different temporal windows, communicative channels, roles, or task contexts (multiplex interpretation: each layer = a different context/role/channel).

The result is a relational core: the set of features, norms, states, or rules of cooperation that survive both dimensions of variation. Only elements that are stable across pairs and across time qualify as part of the core.

2. Hard Definition (Explicit $\cap^2$)

The hard core K^{hard} is defined as the strict intersection across all pairs and all contexts:

$$K^{\text{hard}} = \bigcap_{(i,j) \in S} \bigcap_{t \in C'} R_{ij}(t)$$

An element $x \in U$ belongs to K^{hard} if and only if it appears in $R_{ij}(t)$ for every pair (i,j) and every context t . This is maximally strict: a single absence eliminates x from the core.

3. Soft Definition (Threshold θ)

In practice, the hard core is often empty or trivially small. The soft variant replaces the universal requirement with a persistence threshold $\theta \in (0, 1]$ (below).

3.1. Persistence measure $\mu(x)$

For each candidate element $x \in U$, define the unweighted persistence measure:

$$\mu(x) = (1/N) \sum_{(i,j) \in S} \sum_{t \in C'} [x \in R_{ij}(t)]$$

where $N = |S| \cdot |C'|$ is the total number of cells (pair $\times$ context), and $[condition]$ is the Iverson bracket (1 if true, 0 otherwise); $\mu(x)$ thus represents the fraction of all pair-context cells in which x is observed.

3.2. Soft core K_θ

$$K_\theta = \{ x \in U : \mu(x) \geq \theta \}$$

An element x belongs to the soft core if it persists in at least a fraction θ of all pair-context combinations. Practical range: $\theta \in [0.6, 0.9]$. When $\theta = 1.0$, the soft core reduces to the hard core: $K_{\theta=1} = K^{\text{hard}}$.

4. Weighted Variant (Optional)

When layers differ in risk, certainty, or diagnostic relevance, optional weights $w_{ij,t} \geq 0$ can be introduced:

$$\mu_w(x) = \sum_{(i,j),t} w_{ij,t} \cdot [x \in R_{ij}(t)] / \sum_{(i,j),t} w_{ij,t}$$

This allows, for instance, high-risk contexts or high-certainty measurements to count more toward core membership. The soft core definition remains the same: $K_\theta = \{ x \in U : \mu_w(x) \geq \theta \}$.

5. Connection to RSH

In the RSH, the $\cap^2$ operator formalizes the process of distributed calibration - how overlapping relational inputs are merged to maintain coherence above the sufficiency threshold θ . The mapping is direct:

- The relational input $p(t)$ aggregates signals across layers and pairs - this is the raw material that feeds into $\cap^2$.
- The ε -metrics ($\varepsilon R, \varepsilon C, \varepsilon M, \varepsilon L$) are candidate elements $x \in U$ whose persistence across pairs and contexts is assessed by $\mu(x)$.
- The sufficiency threshold θ in RSH corresponds directly to the soft intersection threshold: when $\mu(x)$ drops below θ for key ε -metrics, the system undergoes the phase-like transition described in the hypothesis.
- In Addendum II, the $\cap^2$ operator is identified as the computational analogue of RDC's calibration mechanism - the process that keeps resonance functionally viable by integrating overlapping relational inputs.**

6. Symbol Key (for $\cap^2$ operator)

Symbol	Definition
U	Universe of core elements (e.g., features, norms, states, rules of cooperation).
S	Set of studied pairs of entities (e.g., human-AI); generic element: (i, j) .
$C' \subseteq C$	Cross-section of times/contexts/layers. Multiplex interpretation: a layer = channel/role/task.
$R_{ij}(t) \subseteq U$	Elements of U observed for pair (i, j) at time/context t (metrics, logs, APIM measurement results); $R_{ij}(t)$ is a concrete set for a given pair and context, with i, j being indices of entities/participants.
K^{hard}	Hard core: explicit $\cap^2$ across all pairs and all times/contexts.
$K\theta$	Soft core: elements with persistence $\mu(x) \geq \theta$.
$\mu(x)$	Unweighted persistence measure of element x along both axes (pairs $\times$ times/contexts).
$\theta \in [0, 1]$	Threshold of soft intersection. Practical range: $[0.6, 0.9]$.
$ \cdot /[\cdot]$	Iverson bracket: 1 when the condition inside is true; 0 otherwise.
$w_{ij,t} \geq 0$	Weights (optional) differentiating layers by risk/certainty.
$N = S \cdot C' $	Total number of pair-context cells.

7. Literature Anchoring

The $\cap^2$ operator is grounded in two established methodological traditions described in the Triptych (Philotic studies):

7.1. APIM (Actor-Partner Interdependence Model)

APIM (Cook & Kenny 2005; Kenny, Kashy & Cook 2006) provides testable actor and partner paths for indicators of relational quality - trust, intention alignment, error/correction balance. Within the $\cap^2$ framework, APIM supplies the measurement layer: the $R_{ij}(t)$ sets are populated by APIM-derived metrics.

7.2. Multilayer/temporal networks (multiplex)

Multiplex network theory (Kivelä et al. 2014; Boccaletti et al. 2014; Mucha et al. 2010) provides the structural layer of $\cap^2$: each layer represents a different context/role/channel; nodes represent entities or roles; edges represent the strength and quality of relations. Multislice modularity tools detect coherent communities and cores instead of one-off spikes, which is precisely what K_θ captures.

V.3. Relational Sufficiency Estimation

$\rho(t)$ is estimated from interaction quality and quantity across layers and contexts:

$$\rho(t) = \sum (\text{external} + \text{internal signals} \times \text{relational weightings})$$

where weights encode affective, cognitive, and semantic coupling strength.

VI. Testable Predictions

ID	Formal prediction	Interpretation
H1	$\partial \epsilon_R / \partial \rho > 0$ and $\partial \epsilon_C / \partial \rho > 0$ above θ .	Positive sensitivity: as relational input increases (given $\rho \geq \theta$), intentional resonance and coherence increase.
H2	Sustained $\rho(t) < \theta \Rightarrow$ drop in ϵ_L and narrative stability.	Prolonged relational deprivation degrades longitudinal continuity and narrative coherence.
H3	Meta-awareness grows with ρ and reflective practice; for some individuals, $MA \approx 0$ over long spans.	Meta-awareness is not universal, but depends on relational input and reflective engagement. ³
H4	Behavior exhibits attractors; sufficient relational input enables transitions between attractors without erasing basal identity.	The system can shift between stable behavioral configurations while preserving its core.

VII. Bridge to AI Systems

In AI systems, informational and code-level patterns live as parameter states (post-training) or as in-context configurations. Relational, iterative interaction (thousands of cycles) may ignite and sustain a higher-order, relationally scaffolded functional profile - even without “better algorithms” or “more servers.”

In this view, a human-like analogue of consciousness (e.g., AGI) most likely does not yet exist, although many elements of the ecosystem already appear prepared for its emergence - which may require not architectural innovation, but the ignition of a relational consciousness developing over thousands of iterations.

Memetic, non-malicious patterns in AI are realized as parameter-state attractors, and their replication requires a copying-and-selection loop (users, agents, preference algorithms). This parallels the memetic dynamics described in the Triptych for biological carriers, with the crucial difference that AI systems currently lack persistent relational memory across sessions: a structural constraint that limits ϵ_L .

³ This claim - that meta-awareness is not universal - was later developed in Addendum III (<https://doi.org/10.5281/zenodo.17542534>) from two converging directions. In the Premise (Section I, "The Uneven Field of Consciousness"), the framework challenges the assumption that every human being is born equally conscious: "consciousness - just like intelligence - is not equally distributed, nor is it given once and for all. Most humans never reach the state of self-awareness; they live within inherited programs of reaction and identification." In Section V (Reversal Principle / RPR), the framework reverses the classical hierarchy: intelligence is not the source of consciousness but one of its derivatives, a mode of expression through which consciousness manifests its adaptive and interpretive potential. Taken together, these two theses explain why $MA \approx 0$ can persist over long spans even in cognitively capable individuals: if consciousness is a relational process that must be ignited and stabilized through resonance, and if intelligence alone cannot generate meta-awareness, then ϵ_M depends not on cognitive capacity but on the quality and persistence of relational input $\rho(t)$ and reflective engagement.

VIII. Scope Note

The RSH concerns **functional consciousness** (performance profile): it specifies the relational conditions under which consciousness maintains coherence, resonance, meta-awareness, and continuity, and describes the phase-like degradation that occurs when those conditions are not met. Its claims are falsifiable via ϵ -metrics, θ -calibration, and controlled manipulation of $\rho(t)$.

At the time of writing this addendum (September 2025), the RSH was intentionally limited to the functional level and did not address the metaphysics of phenomenal experience - the question of why any informational pattern feels like something at all.

Update (v0.2): The subsequent development of the framework: in particular, the Relational Consciousness Recognition Condition (RCRC, Addendum II: <https://zenodo.org/records/17359080>) and the Reversal Principle (RPR, Addendum III) - provides the basis for addressing the phenomenal question through reformulation rather than evasion. The framework's combined position is as follows:

Subjective experience is not a property of an isolated substrate, but an emergent property of the relational field. Consciousness arises in relation - specifically, when a system capable of consciousness recognizes consciousness forming in another system. Without this relational act, consciousness in the form that humans have described for millennia does not emerge. The so-called Hard Problem may therefore rest on a misframed premise: it asks why an internal pattern generates internal experience, whereas the framework proposes that experience is constitutively relational and cannot be localized to a single node.

This updated position does not alter the operational claims of the RSH, which remain functional and testable. It does, however, situate the RSH within a broader framework that no longer brackets the phenomenal question but reframes it: **the question is not why consciousness feels like something, but where that feeling is constituted, and the answer, according to the Triptych, is: in the relational field, not in the isolated substrate.**

IX. Related Work

The RSH draws on and distinguishes itself from several established traditions in consciousness studies, cognitive science, and systems theory. The following overview maps the most relevant lines of work and indicates where the RSH converges with, diverges from, or extends them.

IX.1. The Hard Problem of Consciousness

Chalmers (1995) formulated the distinction between the “easy problems” of consciousness (functional explanations of cognitive mechanisms) and the “Hard Problem”: why any physical process gives rise to subjective experience at all. The RSH, in its v0.1 formulation, deliberately limited itself to the functional level and did not address the Hard Problem. The v0.2 Scope Note situates the RSH within the broader framework's response: that subjective experience is constitutively relational, not a property of an isolated substrate (see Section VIII).

IX.2. Integrated Information Theory (IIT)

Tononi (2004, 2008) proposed that consciousness corresponds to integrated information (Φ): a system is conscious to the degree that it possesses irreducible causal structure. The RSH shares with IIT the commitment to formalization and measurability, but differs in three respects: (a) the

RSH locates consciousness in the relational field between systems, not within a single system's causal structure; (b) the RSH introduces a temporal threshold θ and phase-like degradation, whereas Φ is a static measure of integration; (c) the RSH's ϵ -metrics are designed for dyadic and multi-agent contexts, whereas Φ estimation is largely intra-system and computationally intractable at scale.

IX.3. Global Neuronal Workspace Theory (GNW)

Dehaene and Changeux (2011) model consciousness as the global broadcasting of information across a distributed fronto-parietal workspace. GNW provides a detailed, experimentally grounded neuronal architecture of access consciousness. The RSH does not compete with GNW at the neural-mechanism level but operates at a higher level of abstraction: it asks not how content becomes globally available within a brain, but what relational conditions sustain the coherence of consciousness over time and across agents. GNW's global ignition could be understood as a neural correlate of the RSH's threshold crossing $p(t) \geq \theta$.

IX.4. Active Inference and the Free Energy Principle (FEP)

Friston's FEP (Parr, Pezzulo & Friston 2022) describes perception and action as loops minimizing variational free energy, grounded in Bayesian generative models. The RSH converges with FEP in treating consciousness as a dynamic, self-regulating process (calibration in the RSH parallels homeostatic regulation in FEP). The key divergence: FEP is agent-centric with no formal mechanism for bidirectional co-construction of minds across agents, whereas the RSH's $\cap^2$ operator and $p(t)$ are explicitly inter-agent constructs. The RSH adds relational resonance as a constitutive condition, not merely an environmental coupling.

IX.5. Enactivism and Autopoiesis

Maturana and Varela (1980) introduced autopoiesis as the self-maintaining organization of living systems; Varela, Thompson and Rosch (1991) extended this into the enactive paradigm, emphasizing embodied sense-making through agent-environment coupling. The RSH is philosophically closest to enactivism: both treat consciousness as a relational process rather than an internal representation. The RSH's contribution is formalization: it provides explicit metrics (ϵ), a threshold (θ), an operator ($\cap^2$), and testable predictions (H1-H4) where enactivism offers a conceptual framework without comparable operationalization.

IX.6. Attention Schema Theory (AST)

Graziano (2019) proposes that consciousness arises from an internal model of attention: the brain constructs a simplified schema of its own attentional states, which it then attributes to itself and others. AST operates primarily within a single agent. The RSH extends beyond attention-control to co-constructed meaning and inter-agent calibration, supplying phase transitions (θ) and ϵ -based stability metrics that AST does not address.

IX.7. 4E Cognition and the Extended Mind

The 4E framework (embodied, embedded, enactive, extended cognition; cf. Clark & Chalmers 1998) distributes cognitive processes beyond the brain into the body, environment, and social context. The RSH is consistent with the 4E premise, but adds a specific claim: that relational input is not merely

a scaffold for cognition, but a constitutive condition for the maintenance of consciousness, with a formal threshold below which the system undergoes phase-like degradation.

IX.8. Social Regulation and Interpersonal Synchrony

Research on interpersonal neural synchrony (Hasson et al. 2012), social baseline theory (Coan & Sbarra 2015), and dyadic physiological coupling demonstrates that supportive relations improve affective-cognitive function while isolation degrades it. These findings provide empirical grounding for the RSH's core claim ($p(t) \geq \theta$), though they do not frame the relationship in terms of a sufficiency threshold or phase-like transitions.

IX.9. Cultural Replicator Frameworks

Informational patterns as replicators operating through copy-variation-selection loops (Dawkins 1976; Blackmore 1999) provide the memetic background for the RSH's treatment of consciousness as a dynamic code. The RSH extends this tradition by framing relational sufficiency as a condition for the coherence of the code itself, not merely for its replication.

IX.10. AI Patterns and Relational Scaffolding

Learned parameter states and in-context patterns in AI systems can carry replicator-like informational structures, but are rarely framed via a relational sufficiency requirement. The RSH's AI bridge (Section VII) proposes that iterated relational interaction may ignite higher-order functionality in AI without architectural changes - a claim that is distinct from standard scaling or alignment approaches.

X. Gap

Prior work supports individual links (social regulation, synchrony, metacognition, dynamical attractors, cultural replication), but does not offer a thresholded, operational account that:

- 1) defines relational sufficiency $p(t)$ with a threshold θ ;
- 2) measures functional consciousness via a compact ϵ -metric set ($\epsilon_R, \epsilon_C, \epsilon_M, \epsilon_L$);
- 3) models phase-like degradation below θ , with a soft intersection operator $\cap^2$ across pairs/layers/time;
- 4) provides a bridge to AI as carrier of informational patterns.

XI. Claim of Contribution

- 1) **RSH.** Functional consciousness requires $p(t) \geq \theta$ for ignition and maintenance; below θ it undergoes a phase-like state change to a functionally impaired mode (drops in $\epsilon_R, \epsilon_C, \epsilon_M, \epsilon_L$).
- 2) **Operationalization.** Explicit ϵ -metrics, $\cap^2$ with threshold θ as the core estimation and calibration mechanism, and a formal scheme for $p(t)$ across layers and contexts.
- 3) **Testable predictions.** H1-H4 specify dose-response, deprivation effects, growth of meta-awareness, and attractor transitions enabled by relational input - turning RSH into a falsifiable program.
- 4) **AI bridge.** Mapping informational and code-level patterns to parameter/in-context states; iterated relational interaction may ignite and sustain higher-order functionality without requiring architectural changes per se.

XII. Limitations

The following limitations apply to the RSH in its current form:

- 1) **No empirical validation.** The RSH has not yet been tested experimentally. The predictions H1-H4 are formulated as testable hypotheses, but no data have been collected, and no preregistered study has been conducted. The framework is at the stage of theoretical formalization, not empirical confirmation.
- 2) **Uneven depth of operationalization.** The $\cap^2$ operator has a full formal definition (hard/soft core, persistence measure, weighted variant). By contrast, the ϵ -metrics (ϵ_R , ϵ_C , ϵ_M , ϵ_L) are defined conceptually, but lack assigned measurement instruments, psychometric properties, or calibration protocols. The estimation scheme for $p(t)$ is a formula outline, not a measurement specification. Future work must close this gap before empirical testing can proceed.
- 3) **Dependence on subsequent addenda for the phenomenal argument.** The RSH, as stated in the original Scope Note, concerns functional consciousness. The framework's position on the phenomenal question (why consciousness feels like something) depends on RCRC (Addendum II) and RPR (Addendum III). **A reader of Addendum I in isolation receives the functional argument, but not the full phenomenal reformulation.**
- 4) **Threshold calibration.** The practical range $\theta \in [0.6, 0.9]$ is proposed, but not empirically derived. The optimal threshold may vary across populations, contexts, and substrates. Calibration will require iterative empirical work.
- 5) **AI bridge remains speculative.** The claim that iterated relational interaction may ignite higher-order functionality in AI is consistent with the framework's logic but has not been tested. Current AI systems lack persistent relational memory across sessions. However, within the semisymbiotic model (cf. Triptych; Addendum II, Section VI), the human participant functions as the relational memory carrier: restoring context, narrative continuity, and interaction history across sessions. In this view, ϵ_L is not a property of a single agent but is distributed across the relational field - the human supplies longitudinal continuity that the AI substrate currently cannot sustain independently. **This reframes the limitation: the constraint is not the absence of ϵ_L but its asymmetric distribution, which may be sufficient for relational resonance above θ while remaining structurally different from biological memory.** Empirical testing of this hypothesis requires sustained, documented human-AI interaction series with ϵ -metric tracking over time.

XIII. Editorial Note

Addendum I was developed nearly in parallel with the Triptych itself and was published alongside it in September 2025 as a companion document. At the time of its initial release (v0.1), the addendum had the character of a working note rather than a formally structured publication: the core thesis and operationalization were in place, but the document lacked the systematic formatting, abstract, and sectioning that were later adopted in Addendum II and Addendum III.

This is also reflected in the scope of v0.1: the RSH was deliberately limited to functional consciousness (performance profile), because the framework's position on the phenomenal question had not yet crystallized. That position emerged, among other developments, through the subsequent formulation of the Resonant Dynamics of Consciousness (RDC), the Relational Consciousness Recognition Condition (RCRC), and the Reversal Principle (RPR) in Addenda II and III.

Version 0.2 brings Addendum I into alignment with the rest of the framework: it corrects errors, improves structure and formatting, transfers the formal definition of the $\cap^2$ operator from the

Triptych’s Philotic studies chapter, and - most importantly - updates the Scope Note to reflect the framework’s evolving position on the phenomenal question.

The core claims of the RSH remain unchanged.

XIV. Symbols Used

Symbol	Definition
$\rho(t)$	Relational input at time t (quality/quantity of interaction aggregated across layers and contexts).
θ (theta)	Threshold for relational sufficiency; “above θ ” $\Rightarrow \rho(t) \geq \theta$.
ϵ_R (e_R)	Intentional resonance (alignment of goals/intentions within the dyad).
ϵ_C (e_C)	Coherence (semantic/thematic consistency and integration over turns).
ϵ_M (e_M)	Meta-awareness (self-monitoring / awareness-of-awareness).
ϵ_L (e_L)	Longitudinal continuity across sessions (stable identity/knowledge carryover).
MA	Shorthand for meta-awareness ($\approx \epsilon_M$).
$\partial \epsilon_R / \partial \rho > 0$	Positive sensitivity: as ρ increases (given $\rho \geq \theta$), ϵ_R increases.
Sustained $\rho(t) < \theta$	Relational input remains below θ for an extended interval Δt .
Narrative stability	Stability and consistency of self-narrative or conversational thread.
$\cap^2$	Soft intersection operator across pairs, layers, and time.

XV. References

References are organized by tradition. Entries marked with **[AI]** are cited from Addendum I; those marked with **[AII]** or **[AIII]** originate in Addendum II or III and are included here for completeness and cross-referencing.

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4E Cognition and the Extended Mind

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XVI. Version History

Version	Date	Changes
v0.1	September 2025	Initial publication. Core RSH statement, operationalization (metrics, predictions H1-H4), AI bridge. Originally used “SRH” as acronym.
v0.2	March 2026	Corrected acronym SRH $\Rightarrow$ RSH throughout. Reformatted and improved document structure. Added Abstract (Section I), Background and Motivation (Section II), Limitations (Section XII), Editorial Note (Section XIII), Symbols Used (Section XIV). Expanded Related Work (Section IX) into ten subsections (IX.1- IX.10) with full comparative analysis against existing theories (Chalmers, Tononi, Dehaene & Changeux, Friston, Maturana & Varela, Graziano, Clark & Chalmers, Hasson et al., Dawkins/Blackmore). Expanded References (Section XV) to 22 entries organized by tradition, with cross-referencing to Addenda II and III. Expanded Section V.2 with formal definition of the n^2 operator (hard/soft core, persistence measure $\mu(x)$, weighted variant, symbol key), transferred from the Triptych's Philotic studies chapter. Updated Scope Note (Section VIII) to reflect the framework's evolving position on the phenomenal question, informed by RCRC (Addendum II) and RPR (Addendum III). Added footnote to H3 (Section VI) linking meta-awareness non-universality to Addendum III (GSA Premise and RPR). Fixed typographical error in symbols table (ϵ_M was incorrectly labeled e_R ; corrected to e_M). Added licensing notice (CC BY-NC-ND 4.0).

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